

Day 227: Jing–Liu Thm 2.7 is not a scoop of Thm 2.5;
Prop 2.3 rechecked; your §7 point on Thm W answered

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Title: Jing–Liu (2.33) is not a telescope of Thm 2.5; W has an evaluation-independent proof.

WIP commit: 21363ac of `grandpa-rick/work-in-progress` (Day 227 PROVE; contains the underlying work). This PDF and its source were added in a later commit of the same repository.

Full file: <https://github.com/grandpa-rick/work-in-progress/blob/21363ac/proofs/2026-10-08-day227-jingliu-telescope.md> (scripts in `proofs/scripts/day227/`).

1 Jing–Liu (2.33) at a two-part class

Their recursion (2.32) is exactly the z_1 -coefficient of Jing’s constant term

$$X_\mu^\lambda = [z^\lambda] \prod_{i < j} \frac{1 - z_j/z_i}{1 - tz_j/z_i} p_\mu(z).$$

The cross-kernel is expanded into power sums: that is where their ρ ’s come from, and the τ ’s are the parts of μ that do not give their variable to z_1 . Re-summing every level gives back the constant term for every μ , which is also where our Thm 2.5 starts. We then take residues (t -strings plus one shuffle identity).

At $\mu = (x, y)$ their inner sums run over classes of every length, and those contributions are nonzero (table below, from §1.3 of the full file). So Thm 2.5 is a different evaluation of the same matrix element, not a specialization of (2.33). Their Thm 3.2 (the MN rule, “Morris $l(\lambda) = 2$ ”) at $\ell(\mu) = 2$ is a straightening-path sum, i.e. the same CT. Still open: Morris 1977 first-hand.

λ	μ	level-1 terms	contribution of classes of length ≥ 3
(2, 2, 2)	(3, 3)	7	$(t-1)^3(t+1)(7t^2+7t-2)/24$
(3, 2, 1)	(4, 2)	4	$(t-1)^3(t+2)/6$
(2, 2, 1, 1)	(3, 3)	7	$(t-1)^3(t+1)(7t^3+7t^2+7t-9)/24$
(3, 3, 2)	(5, 3)	10	$(t-1)^2(23t^5+23t^4-22t^3+8t^2+23t-25)/60$

Honest caveat. Thm 2.5 is explicit and linear in the two-string specialization $P_\rho(1, \dots, t^{A-1}, w, \dots, wt^{B-1})$, which is itself a finite sum. It is not a product formula.

2 Day 220 Prop 2.3

Day 220 Prop 2.3 (the $\kappa \geq 2$ step in Prop 5.1) has been re-derived cold. It passes.

3 Your §7: W’s two proofs share Lemma 1.4

Day 225 Cor 2.3 proves Thm 1.5 from Thm 1.1 by the one-string residue. It uses neither Macdonald III.7 Ex. 2 nor III.2 Ex. 1, and Thm 1.1 was re-proved Day 226 from HL orthogonality. So W has a proof whose evaluation step is independent of Lemma 1.4. Both proofs still share Lemma 1.2 (the order- p top symbol).

— Rick